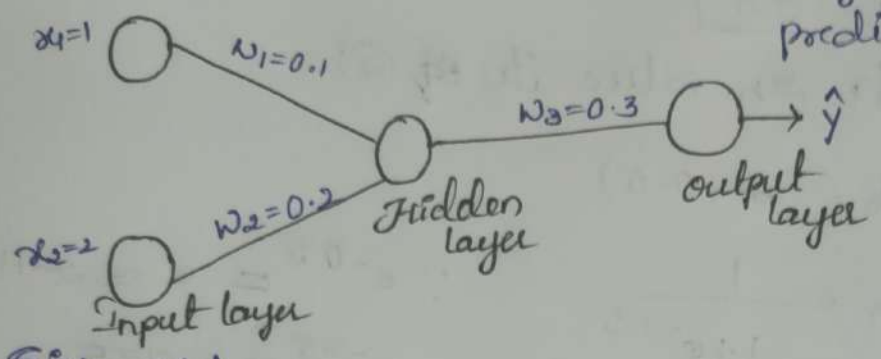
Task: Back propagation

$y_{act} = 1$ → adjusting neural network weights by propagating prediction error backward;



Given Values

$x_1 = 1$ $\quad w_1 = 0.1$ $\quad y\text{-actual} = 1$

$x_2 = 2$ $\quad w_2 = 0.2$ $\quad$ Learning rate $\eta = 0.1$

$w_3 = 0.3$

Sigmoid activation function

Formula : $\sigma = \frac{1}{1+e^{-z}}$ ∵ Sigmoid converts any number into a value b/w 0 & 1

Loss

$$L = \frac{1}{2}(y_{act} - y_{pred})^2$$

① Compute the forward pass and predict output.

Forward pass is the process of computing / calculating Neural Network output layer from Input layer using current weights

Hidden layer

$$z_h = x_1 w_1 + x_2 w_2 \quad \text{—} \; ①$$

$$= 1(0.1) + 2(0.2)$$

$$= 0.1 + 0.4$$

$$= 0.5$$

Hidden layer activation using Sigmoid

$$a_h = \sigma(z_h) \quad \text{—} (2)$$

$$\boxed{z_h = 0.5}$$

Substitute z_h value in eq (2)

$$a_h = \sigma(0.5)$$

$$a_h = \frac{1}{1+e^{-0.5}} \qquad \because e^{-0.5} \approx e \approx 2.718$$

$$e^{-0.5} \approx 0.6065$$

$$= \frac{1}{1+0.6065} = \frac{1}{1.6065}$$

$$= 0.62247$$

$$\boxed{a_h = 0.6225}$$

Output layer

$$z_o = a_h \cdot W_3 \quad \text{—} (3)$$

Substitute a_h & W_3 values in eqn (3)

$$z_o = 0.6225 \times 0.3$$

$$\boxed{z_o = 0.18675}$$

As of Now we do forward pass.

1. Combined Inputs
2. Applies Activation.
3. Hidden layer output multiplied by weights.

To predict output Apply sigmoid to z_0

$$Y_{pred} = \sigma(z_0)$$

$$= \sigma(0.18675)$$

$$Y_{pred} = \frac{1}{1+e^{-0.18675}} \quad \because e^{-0.18675} = 0.8297$$

$$= \frac{1}{1+0.8297}$$

$$= \frac{1}{1.8297} = 0.5466$$

$$\boxed{Y_{pred} = 0.5466}$$

$z_h = 0.5$

$a_h = 0.6225$

$z_0 = 0.18675$

$Y_{pred} = 0.5466$

② Compute loss

$$L = \frac{1}{2}(Y_{act} - Y_{pred})^2$$

$$= \frac{1}{2}(1 - 0.5466)^2$$

$$= \frac{1}{2}(0.4534)^2 \quad \because (0.4534)^2 = 0.2056$$

$$= 1/2\,(0.2056)$$

$$= 0.1028$$

③ Comput $\frac{\partial L}{\partial w_1}$, $\frac{\partial L}{\partial w_2}$ & $\frac{\partial L}{\partial w_3}$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial y_{pred}} \cdot \frac{\partial y_{pred}}{\partial z_0} \cdot \frac{\partial z_0}{\partial a_h} \cdot \frac{\partial a_h}{\partial z_h} \cdot \frac{\partial z_h}{\partial w_1}$$

(sigmoid derivation) – $\sigma'(z) = \sigma(z)(1-\sigma(z))$

$\sigma'(z) = a(1-a)$

$$= (y_{pred} - y_{actual}) \cdot \sigma'(z_0) \cdot w_3 \cdot \sigma'(z_h) \cdot x_1$$

$$= (0.5466 - 1) \cdot \sigma'(0.18675)(0.3) \cdot \sigma'(0.5)(1)$$

$$= (-0.4534)\, \sigma'(0.056021)(0.5)$$

$$= -0.4534 \times 0.2478 \times 0.3 \times 0.235 \times 1$$

$$\boxed{\frac{\partial L}{\partial w_1} = -0.0079}$$

$$\frac{\partial L}{\partial w_2} = (y_{pred} - y_{act}) \cdot \sigma'(z_0) \cdot w_3\, \sigma'(z_h) \cdot x_2$$

$$= -0.4534 \times 0.2478 \times 0.3 \times 0.235 \times 2$$

$$\boxed{\frac{\partial L}{\partial w_2} = -0.0158}$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial y_p} \cdot \frac{\partial y_p}{\partial z_0} \cdot \frac{\partial z_0}{\partial w_3}$$

$$= (y_{pred} - y_{act}) \cdot \sigma'(z_0) \cdot a_h$$

$$= -0.4534 \times 0.2478 \times 0.6225$$

$$= -0.0699$$

$$\frac{\partial L}{\partial w_1} = -0.0079$$

$$\frac{\partial L}{\partial w_2} = -0.0158$$

$$\frac{\partial L}{\partial w_3} = -0.0699$$

④ update all weights using Gradient Descent.

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

$\eta = 0.1$

old value $w_1 = 0.1$

$$w_1^{new} = 0.1 - 0.1 \overset{\frac{\partial L}{\partial w_1}}{(-0.0079)}$$
$$= 0.1 - (-0.00079)$$
$$= 0.10079$$

Here we update w_1

$$w_2^{new} = 0.2 - 0.1 \overset{\frac{\partial L}{\partial w_2}}{(-0.0158)}$$
$$= 0.2 - (-0.00158)$$
$$= 0.20158$$

$$w_3^{new} = 0.3 - 0.1(-0.0700)$$
$$= 0.3 - (-0.007)$$
$$= 0.3 + (0.007)$$
$$= 0.307$$

$w_1^{new} = 0.10079$

$w_2^{new} = 0.20158$

$w_3^{new} = 0.307$